
DIGITAL
FORENSICS
PROCESS

DIGITAL FORENSIC PROCESS

CORE PHASES OF THE DIGITAL FORENSIC PROCESS



LOCARD'S EXCHANGE PRINCIPLE

Anyone or anything entering a crime scene takes something with them and leaves something behind.

For example: walking across a field of fresh snow will leave footprints behind and particles of snow on your shoes.

Same is true of digital devices.

For example: reading a text file on a system will produce evidence showing that an application (such as Notepad) was run in order to open the file, and evidence of file access will be produced showing which file was opened, when, where it resides, and by which user account on the system.

CHAIN OF CUSTODY

Tracks the movement of evidence through the entire digital forensic process

Documents each person who handled the evidence, including relevant dates and times, and the purpose of any transfers between people or into or out of storage locations



Chain of Custody Form

Property/Record Number: _____

Anywhere Police Department

EVIDENCE CHAIN OF CUSTODY TRACKING FORM

Case Number: _____ Offense: _____
Submitting Officer: (Name/ID#) _____
Victim: _____
Suspect: _____
Date/Time Seized: _____ Location of Seizure: _____

Description of Evidence		
Item #	Quantity	Description of Item (Model, Serial #, Condition, Marks, Scratches)

Chain of Custody				
Item #	Date/Time	Released by (Signature & ID#)	Received by (Signature & ID#)	Comments/Location

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DELETING A FILE

> When a file is deleted (not put into the Recycle Bin/Trash), file system metadata for the file is marked as available for reuse and the clusters allocated to that file are marked as unallocated.
» Over time, when the OS needs the space, it will overwrite this data.

> While neither the file system metadata nor the file's content has been overwritten, the file system metadata can be used to locate the file and recover it in its entirety, including the original name and timestamps.

DELETING A FILE

> If the file system metadata for the file is overwritten but the file's content is not, the file can potentially be recovered, but its original name and location will not be known. The process of recovering such data is known as file carving.

> If the file's content is overwritten, but the file system metadata for the file is not, it will be possible to see that the file previously existed and potentially see its location, size and timestamps, but not possible to recover it. Note that other evidence relating to a previously existing file may also reside on the system.

ORDER OF VOLATILITY

The order in which digital evidence should be collected to minimize the loss of data.

1. CPU registers and cache, if you have the capability
» This is extremely difficult and would involve specialist hardware and knowledge; in all of our real-world experience, it is highly unlikely that this will be possible

2. Target system RAM

3. Router data

ORDER OF VOLATILITY

4. Target system hard drive

5. Data stored remotely

6. Network topology and configuration

7. Offsite backup data

WRITE BLOCKING

Write blockers can be either hardware devices or software-based. NIST describes the critical requirements of a write blocker to be the following:

> The tool shall not allow a protected disk drive to be changed
> The tool shall allow any information to be obtained from or about any disk drive
> The tool shall allow any operations to a disk drive that is not protected

WRITE BLOCKERS

> Portable device to facilitate read-only access to physical evidence

> Need to be connected to workstation that runs acquisition software to acquire a device

> Required for each type of interface you need to acquire: USB, SATA, NVMe, PCIe

> Validate before use & periodically to check it is actually blocking writes!



TOOLS AND TECHNIQUES

Forensic Workstation(s) (Server/desktop/laptop)



> High-end system with a fast processor, lots of RAM and a lot of storage

> Laptop required for field work

> A number of companies do workstations specifically designed for forensic analysts

DISK DUPLICATORS

> Standalone imaging device with built-in write blocker & screen





> Capability to:

» Image or clone input drive and write to output drive

» Wipe output drive (ready to receive new image)

FARADAY BAG

A Faraday bag is a shielded pouch designed to block electromagnetic signals such as Wi-Fi, GPS, cellular, and Bluetooth. Named after the Faraday cage principle, it uses layers of conductive material to create an enclosure that prevents external signals from penetrating or internal signals from escaping.



SOURCES OF DIGITAL EVIDENCE

Desktop computers and laptops: Typically running one of four operating systems (OS): Windows, macOS, ChromeOS, or Linux ›Will contain artifacts specific to the OS and to installed applications ›Newer devices employ encryption by default

Servers: In addition to standard host-based artifacts, different types of other data are stored on different types of server:

TYPE OF SERVER	PURPOSE	TYPES OF DATA THAT MAY BE STORED
Domain Controller	Authentication of user logins to domain	User information, events such as user logon/logoff
Web Server	Hosts webpages	Access logs, including IP addresses
Proxy Server	Intermediary between user systems and the Internet	Log of requests, configuration data
DHCP Server	Assigns IP addresses and network parameters	Log of assignments
Firewall	Monitors and controls incoming and outgoing network connections	Log of network connection attempts, configuration data
Network-Attached Storage	Stores files and folders	Files, log of file accesses
Terminal Server	Manages authentication to remote applications and systems	Log of remote connection attempts, user account and system information
Application Server	Hosts applications and services on a network	Log of application access
Mail Server	Stores and transmits emails	Audit logs, logs and content for emails sent and received by users

VIRTUAL MACHINES (VMS)

> Typically running one of two Operating Systems: Windows, or Linux

> May be stored locally on a system or in the cloud

> Virtual representation of an entire system, so will contain the same types of artifacts as a physical system of that type, relating to both the OS and applications

WHAT ARE LOGS?

Logs are records of events or activities generated by devices, systems, and applications.

Importance of Logs in Incident Response:

- Provide a timeline of events.
- Help in identifying anomalies and unauthorized activities.
- Essential for forensic investigations.

TYPES OF LOGS

- 1. System Logs:
  - Generated by operating systems (e.g., Windows Event Logs, Linux Syslogs).
- 2. Application Logs:
  - Recorded by software applications (e.g., web servers, databases).
- 3. Network Logs:
  - Generated by network devices (e.g., firewalls, routers, IDS/IPS).

TYPES OF LOGS

4. Security Logs:
   - Captures security events (e.g., authentication attempts, malware detection).
5. Audit Logs:
   - Records changes in configurations and system states.
6. Event Logs:
   - High-level events such as logins, errors, and warnings.

LOG COLLECTION AND MANAGEMENT

- **Centralized Logging:**
  - Collect logs from multiple sources into a central repository.

- **Log Collection Methods:**
  - Syslog protocol
  - Agent-based collection (e.g., Wazuh, Splunk forwarders)

LOG COLLECTION AND MANAGEMENT

- Best Practices:
- Enable logging on all critical systems.
- Use timestamp synchronization (NTP).
- Store logs securely to prevent tampering.

LOG ANALYSIS TECHNIQUES

1. **Identify Baselines:**
   - Understand normal system behavior.

2. **Pattern Recognition:**
   - Look for anomalies or suspicious patterns.

3. **Correlation:**
   - Relate events across multiple logs.

LOG ANALYSIS TECHNIQUES

4. Keyword Search:

- Use specific keywords to identify relevant entries (e.g., "unauthorized access").

5. Trend Analysis:

- Identify trends over time.

6. Use Cases:

- Investigating failed login attempts.
- Tracing the source of a malware infection.

TOOLS FOR LOG ANALYSIS

1. SIEM Tools:
   - Splunk, IBM QRadar, Elastic Stack.

2. Open-Source Tools:
   - Wazuh, Graylog, Logstash.

3. Command-Line Tools:
   - grep, awk, sed, PowerShell cmdlets.

TOOLS FOR LOG ANALYSIS

- 4. Cloud-Based Solutions:
  - AWS CloudTrail, Azure Monitor.

- Key Features:
  - Search and filter logs.
  - Real-time monitoring.
  - Visualizations and dashboards.

THANK YOU!

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